

cost. As a means to protect the fund from these higher liquidation costs, managers may place a maximum level on the position size, such as no more than 10% of ADV.

The effect of transaction costs on portfolio returns and their drag on overall performance has been well documented. For example, Loeb (1983) found that block trading could result in an additional 1–2% of cost for large cap stocks and as much as 17–25% or more for small cap illiquid stocks. Wagner and Edwards (1993) found that implementation of trading decisions could approach almost 3% of the trade value in times of adverse market movement. Chan and Lakonishok (1995) found that the hidden trading cost components due to market impact and opportunity cost could amount to more than 1.5% of trade value. Grinold and Kahn (2000) examined the effect of transaction costs on portfolio construction. Kissell and Malamut (2006) found that inefficient executions, i.e., implementing via strategies or algorithms that are not consistent with the investment objective, can increase tracking error by 10–25 bp for passive index managers and by as much as 50 bp for actively managed funds. And more recently, studies have found that transaction costs may still account for additional slippage of up to 1% of annual performance. With such high trading costs associated with implementation it is no wonder that portfolio managers underperform their benchmarks (Treynor, 1981).

In addition to the high transaction costs and corresponding trading friction, there is often an additional drag on portfolio returns due to a misalignment between the investment objective and trading desk goals. For example, suppose a value manager enters a buy order for a stock that is undervalued in the market. This manager wants to execute the position in an aggressive manner before the market discovers the mispricing and makes a correction. If this order is executed by a trader via a full-day VWAP strategy it is very likely that the market correction will occur before the order is complete causing the manager to pay higher prices or potentially not complete the order fully, which results in high opportunity cost. In either case, the manager does not achieve the full potential of the opportunity because of the trading strategy not because of the investment decision. In this case it would be much more advantageous to trade via a more aggressive strategy, such as an arrival price or implementation shortfall algorithm, in order to transact more shares at the manager's decision price. VWAP strategy in this example is not an appropriate strategy and is misaligned with the investment objective of the fund. Even if the trader achieved or outperformed the VWAP price in this example the selection of the VWAP strategy would be an inappropriate decision.